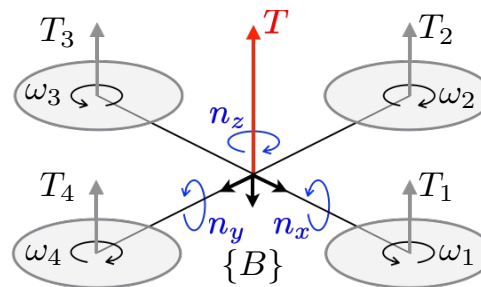


Chapter 2

Quadrotor Dynamic Modeling

UAVs

MEAer - Autumn Semestre – 2022/2023



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Quadrotor Dynamic Modeling

- **Goal:** derive the equations of motion for a quadrotor vehicle
- **Starting point:** rigid-body equations of motion

- Kinematics

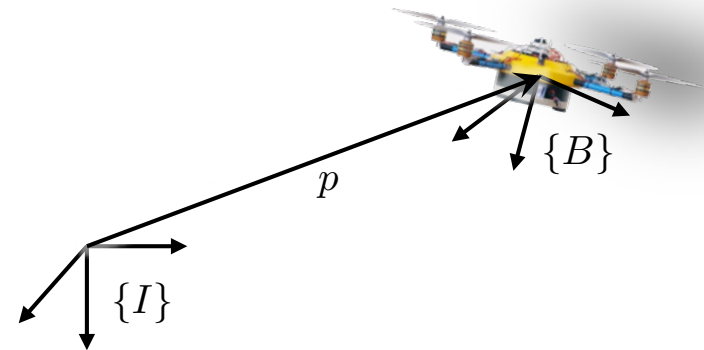
$$\dot{p} = Rv$$

$$\dot{R} = RS(\omega)$$

- Dynamics

$$m\dot{v} = -S(\omega)mv + f$$

$$J\dot{\omega} = -S(\omega)J\omega + n$$



- **1st question:** what is the input?

$$u = ?$$

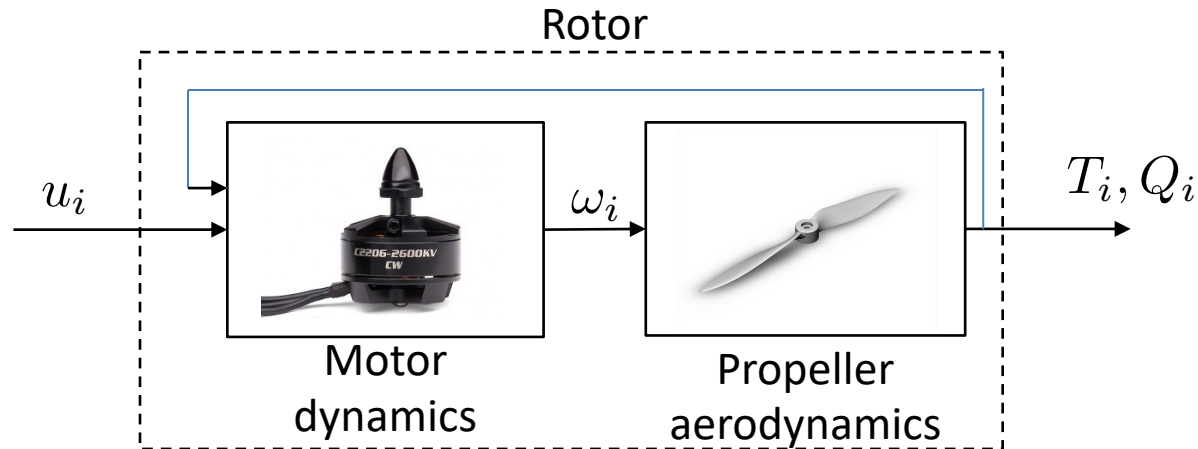
- **2nd question:** what are the generated forces and moments?

$$f = f(p, R, v, \omega, u) = ?$$

$$n = n(p, R, v, \omega, u) = ?$$

Quadrotor actuation – the inputs

- Four rotors: electric motor + propeller



- Input: voltage u_i
- Output: generated thrust and reaction torque T_i, Q_i

Motor Dynamics

- DC Motor Model
 - Mechanical part



$$I \frac{d\omega_i}{dt}(t) = k_t i_i(t) - \tau_i^{load}(t)$$

Load torque

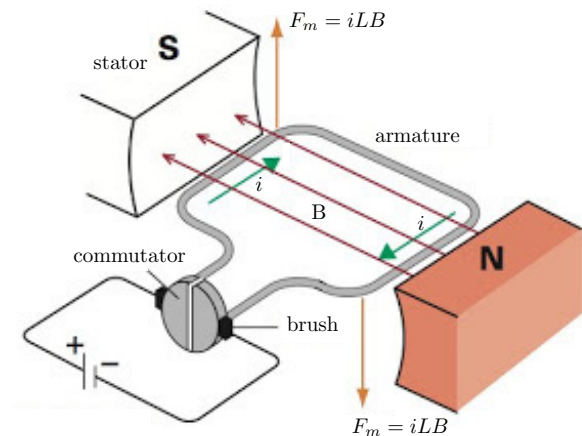
Generated torque:
proportional to current
flowing through the coil

- Electrical part

$$L \frac{di_i}{dt}(t) = u_i(t) - Ri_i(t) - k_b \omega_i(t)$$

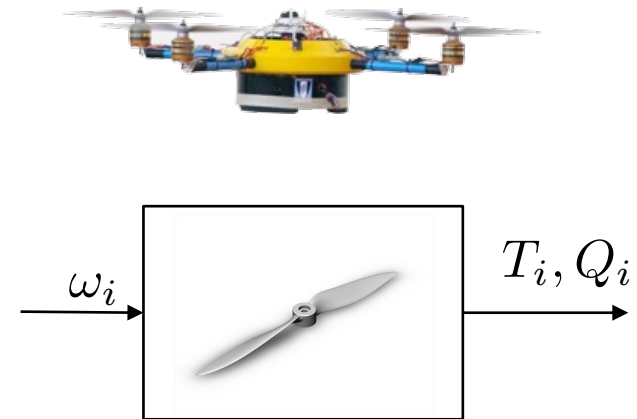
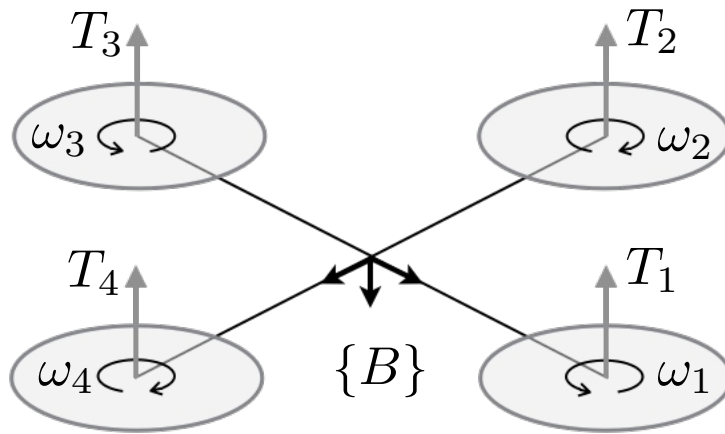
Back electromotive force: proportional
to propeller spinning speed

Input voltage



Quadrotor actuation – propellers

- Two pairs of counter-rotating rotors



- Propellers: thrust and reaction torque
 - Approximately proportional to square of spinning speed

$$T_i = c_T \omega_i^2$$

$$Q_i = c_Q \omega_i^2 (-1)^{i+1}$$

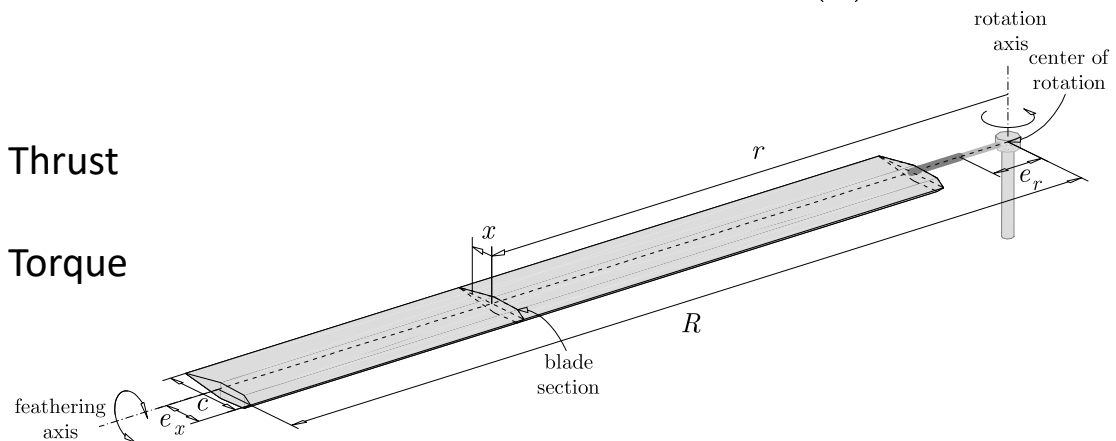
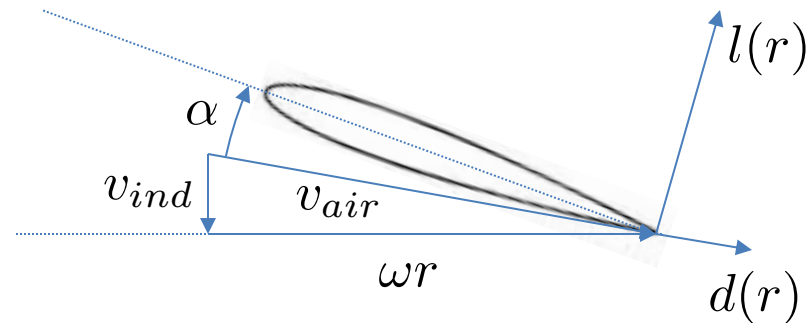
Propellers: Lift and Drag at Blade Element

Lift and Drag at Blade Element

$$l(r) = \frac{\rho}{2} v_{air}^2 c c_l$$

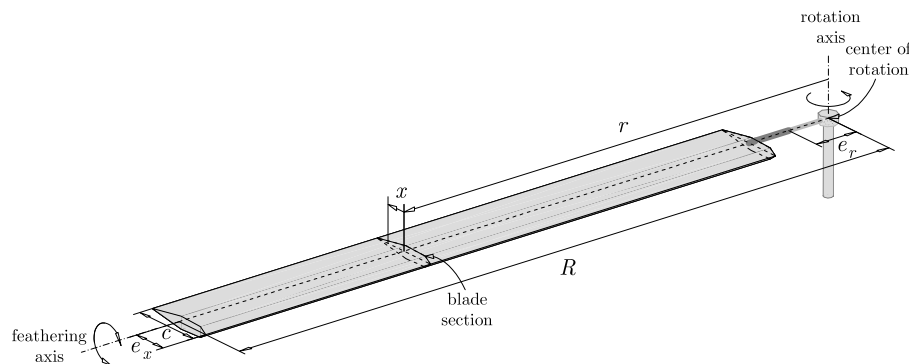
$$d(r) = \frac{\rho}{2} v_{air}^2 c c_d$$

- ρ - air density
- $v_{air} = [\omega_i r \quad v_{ind}]^T$
- c - chord
- $c_l \approx a_0 \alpha$ - lift coefficient $\rightarrow$ Thrust
- c_d - drag coefficient $\rightarrow$ Torque



Propeller Thrust and Reaction Torque

- Integrate lift and drag components over r
- Sum over number of blades N_b



$$T_i = \sum_{j=1}^{N_b} \int_0^R l_j dr = N_b \frac{\rho}{2} R^3 a_0 \left(\frac{\theta}{3} - \frac{v_{ind}}{2} \right) \omega_i^2 = c_T \omega_i^2$$

$$Q_i = \sum_{j=1}^{N_b} \int_0^R (-l_j \alpha_j + d_j) r dr = c_Q \omega_i^2$$

- Thrust and torque constants can be identified experimentally
 - static thrust tests with different payloads

Propeller data example

- APC: https://www.apcprop.com/files/PER2_STATIC-2.DAT

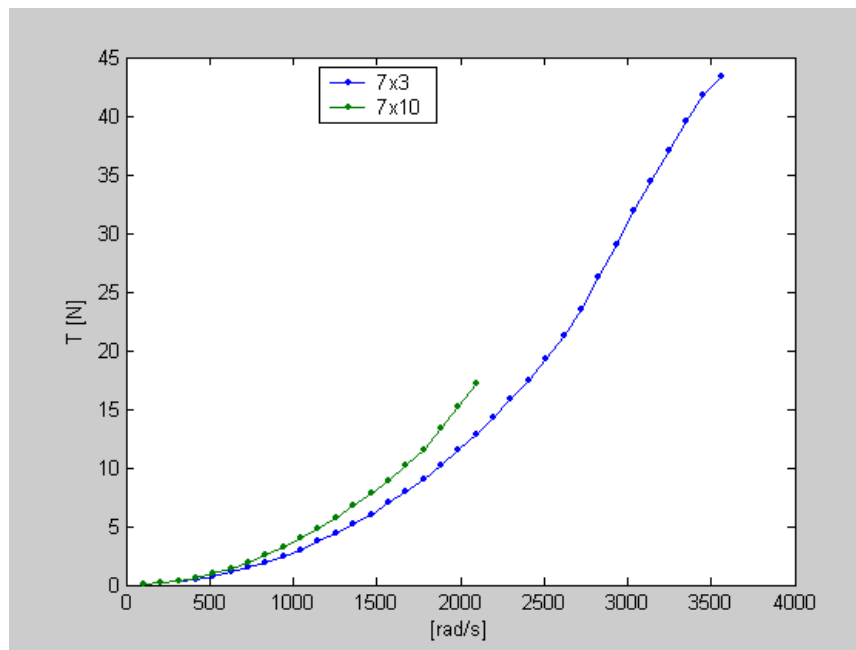
7x10.dat

RPM	THRUST (LBF)	POWER (HP)	TORQUE (IN-LBF)	C _p	C _t
1000	0.01	0.00	0.01	0.09	0.12
2000	0.04	0.00	0.03	0.09	0.12
3000	0.08	0.00	0.06	0.09	0.12
4000	0.14	0.01	0.12	0.09	0.12
5000	0.22	0.01	0.18	0.09	0.12
6000	0.32	0.03	0.26	0.09	0.12
7000	0.44	0.04	0.36	0.09	0.12
8000	0.58	0.06	0.47	0.09	0.12
9000	0.73	0.09	0.60	0.09	0.12
10000	0.90	0.12	0.73	0.09	0.12
11000	1.09	0.15	0.85	0.08	0.12
12000	1.29	0.19	0.97	0.08	0.12
13000	1.51	0.23	1.10	0.08	0.12
13999	1.75	0.27	1.23	0.07	0.12
15000	2.00	0.32	1.36	0.07	0.12
16000	2.28	0.38	1.50	0.07	0.12
17000	2.60	0.58	2.14	0.09	0.12
18000	2.99	0.94	3.31	0.12	0.12
19000	3.41	1.41	4.69	0.15	0.12
20000	3.87	2.01	6.32	0.19	0.13

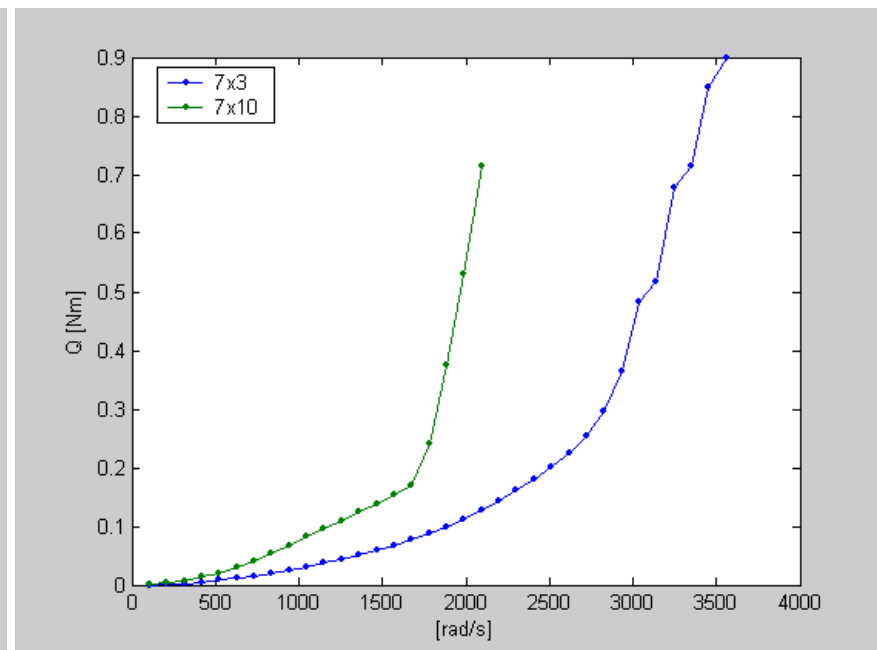
Propeller data example

- APC example – increasing spinning speed
 - Static 7"x3" and 7"x10"

thrust



torque

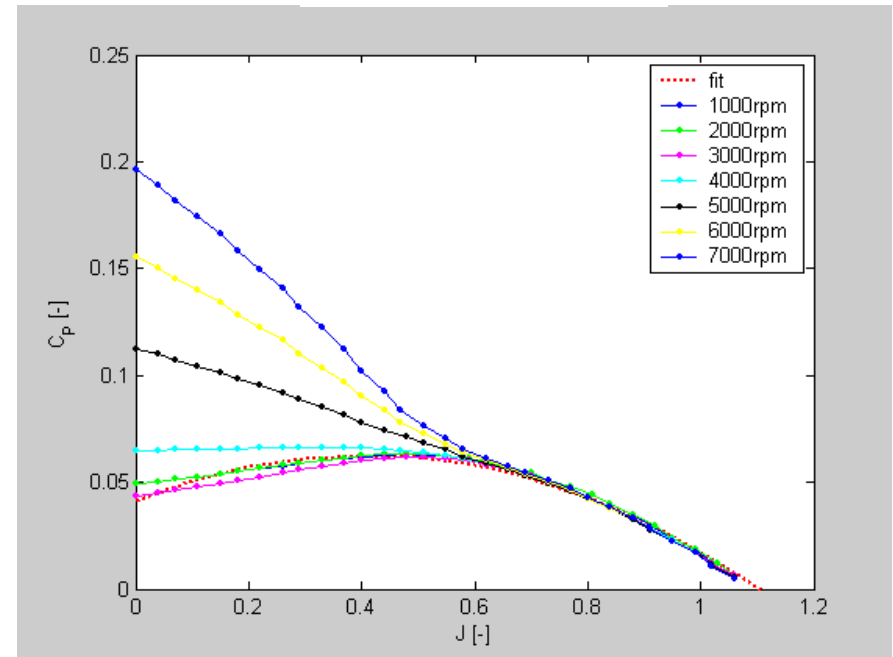
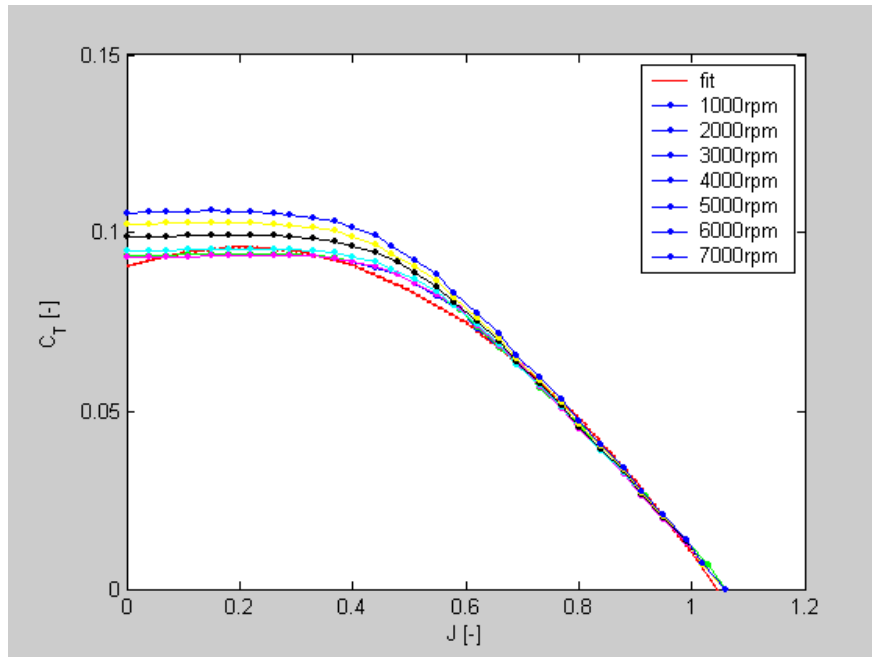


Propeller data when increasing airspeed

- APC 17x13N example – increasing airspeed
 - Considering forward speed and advance ratio: $J = \frac{V}{nD} = \frac{2\pi V}{\omega D}$

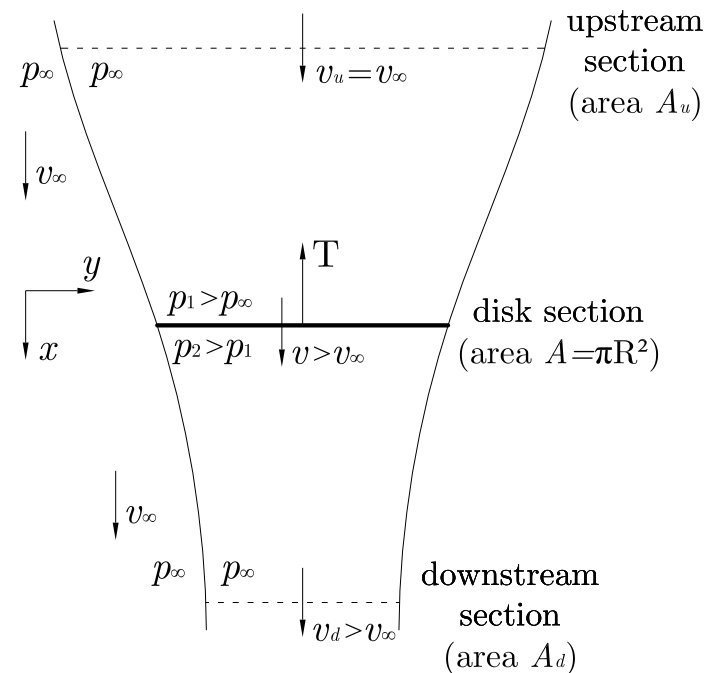
$$C_T = \frac{T}{\rho n^2 D^4}$$

$$C_P = \frac{P}{\rho n^3 D^5}$$



Propeller: further neglected effects

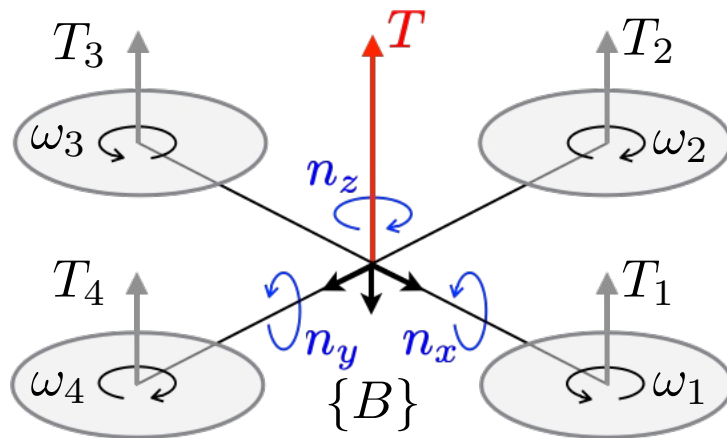
- Each propeller drives a tube of air flow



→ ground effect at low height

Quadrotor input forces and moments

- Two pairs of counter-rotating rotors



$$T = \sum_i T_i$$

$$n_x = l(T_2 - T_4)$$

$$n_y = l(T_1 - T_3)$$

Negligible air velocity

$$\begin{bmatrix} T \\ n_x \\ n_y \\ n_z \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & l & 0 & -l \\ l & 0 & -l & 0 \\ c & -c & c & -c \end{bmatrix}}_{\mathbf{M}} \begin{bmatrix} T_1 \\ T_2 \\ T_3 \\ T_4 \end{bmatrix}$$

$$T_i = C_T \omega_i^2$$

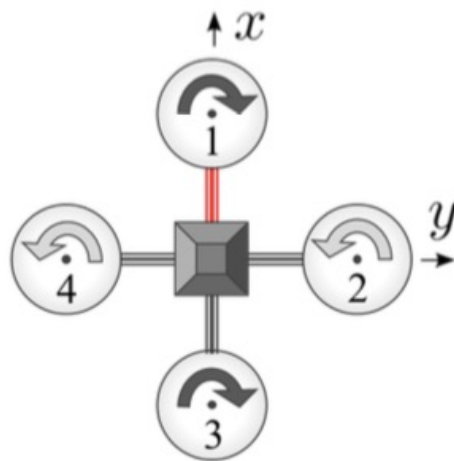
$$Q_i = (-1)^{i+1} C_Q \omega_i^2$$

$$Q_i = (-1)^{i+1} \frac{C_Q}{C_T} T_i$$

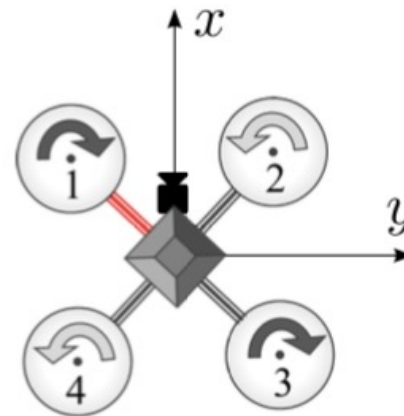
$$n_z = \frac{C_Q}{C_T} (T_1 - T_2 + T_3 - T_4)$$

Quadrotor input forces and moments

- Two pairs of counter-rotating rotors: **exercise**



(a) + configuration



(b) × configuration

- How to express (T, n) as a function of T_i or ω_i : **M** ?
- And if it were an hexa-rotor ?
 - 6 inputs... But **M**= ?

Quadrotor input forces and moments

- Thrust T and moments n_x , n_y , and n_z expressed in $\{B\}$
- Rigid-body dynamics expressed in $\{B\}$
- Near hover: **negligible aerodynamic forces/moments**

$$\begin{aligned}
 m\dot{v} &= -S(\omega)mv + f & f &= -Te_3 + mgR^T e_3 \\
 J\dot{\omega} &= -S(\omega)J\omega + n & &= \begin{bmatrix} 0 \\ 0 \\ -T \end{bmatrix} + \frac{B}{I}R \begin{bmatrix} 0 \\ 0 \\ mg \end{bmatrix} & n &= \begin{bmatrix} n_x \\ n_y \\ n_z \end{bmatrix}
 \end{aligned}$$

- **Exercise:** express the translational dynamics in $\{I\}$

$$m\ddot{p} = ?$$

Quadrotor dynamic model

- Translational dynamics

$$\begin{aligned} \dot{p} &= Rv \\ m\dot{v} &= -S(\omega)mv - Te_3 + mgR^T e_3 \end{aligned} \quad \text{or} \quad m\ddot{p} = -TRe_3 + mge_3$$

- Rotational dynamics

$$\begin{aligned} \dot{R} &= RS(\omega) \\ J\dot{\omega} &= -S(\omega)J\omega + n \end{aligned}$$

- Compact form

$$\begin{aligned} \dot{x} &= f(x, u) & x &= (p, v, R, \omega) \\ u &= (T, n_x, n_y, n_z) \end{aligned}$$

Underactuated System:
 6 DoFs, 4 inputs

Quadrotor dynamic model

- Compact form, as a system

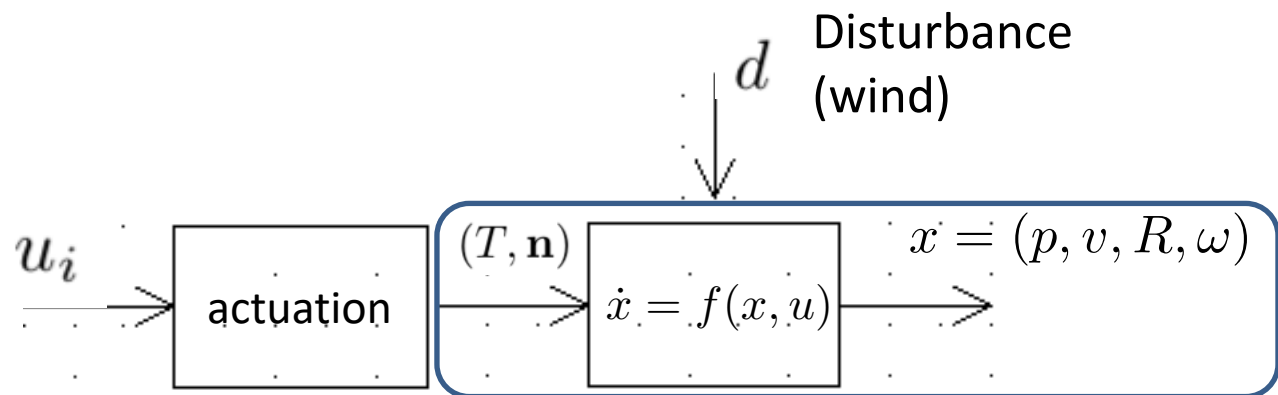
$$\dot{x} = f(x, u) \quad x = (p, v, R, \omega)$$

$$u = (T, n_x, n_y, n_z)$$

Underactuated System:

6 DoFs, 4 inputs

No direct force along x or y



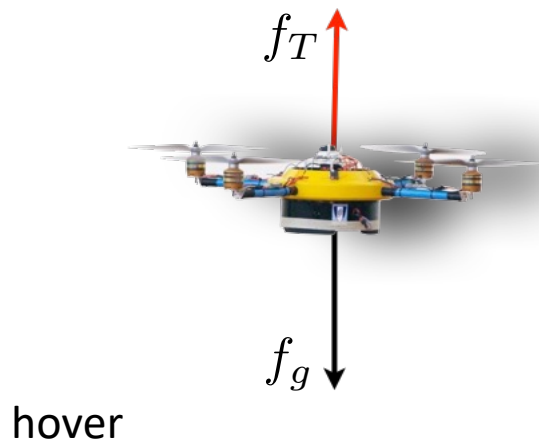
- This is a model for control design
 - No wind, no aerodynamics, ideal actuation: slow flight

Quadrotor dynamic model

- Underactuated system
 - 6 DoFs, 4 independent inputs
 - What trajectories are feasible? We can only prescribe 4 DoFs.

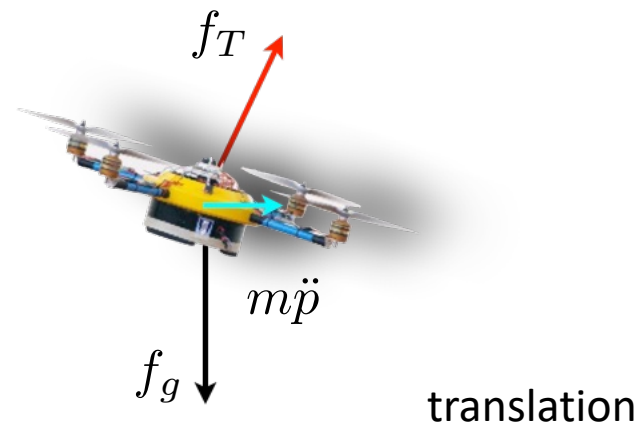
$$m\ddot{\mathbf{p}} = \mathbf{f}_T + \mathbf{f}_g$$

$$\mathbf{f}_T = -TR\mathbf{e}_3 \quad \mathbf{f}_g = m\mathbf{g}\mathbf{e}_3$$



$$m\ddot{\mathbf{p}} = 0$$

$$\mathbf{f}_T + \mathbf{f}_g = 0$$

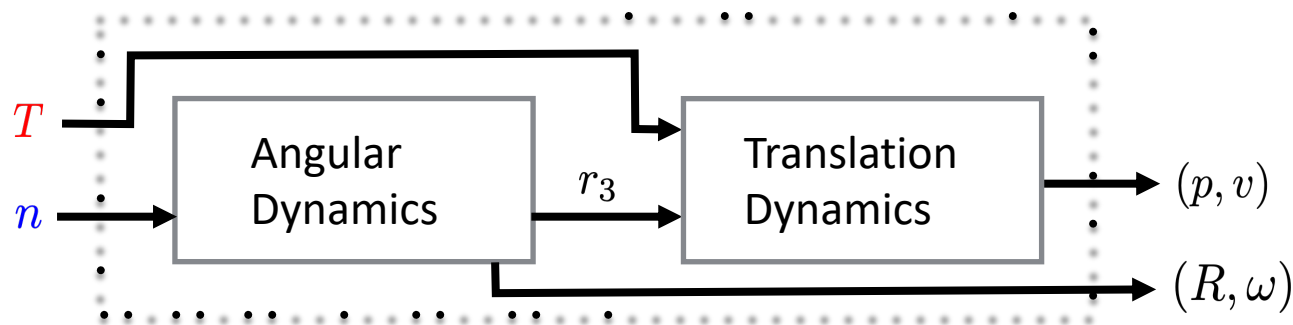


$$m\ddot{\mathbf{p}} \neq 0$$

$$\mathbf{f}_T + \mathbf{f}_g = m\ddot{\mathbf{p}}$$

Quadrotor dynamic model

- Underactuated system
 - 6 DoFs, 4 independent inputs
 - What trajectories are feasible? *We can only prescribe 4 DoFs.*



- We may choose: $y = (p, \psi) \in \mathbb{R}^4$

The quadrotor as a flat system

- What is a flat system?

- State and inputs $(x(t), u(t))$ are completely characterized by an output and its time derivatives

$$(x(t), u(t)) = f(y(t), \dot{y}(t), \ddot{y}(t), \dots)$$

- The *flat* output has the same size of the input

$$u(t) \in \mathbb{R}^m \Rightarrow y(t) \in \mathbb{R}^m$$

- Flat output for a quadrotor

$$y = (p, \psi) \in \mathbb{R}^4$$

- Recall that $x = (p, v, R, \omega)$ $u = (T, n_x, n_y, n_z)$

- It can be shown that $(x, u) = f(p, \dot{p}, \ddot{p}, p^{(3)}, p^{(4)}, \psi, \dot{\psi}, \ddot{\psi})$

The quadrotor as a flat system - Exercise

- Flat output for a quadrotor

$$y = (p, \psi) \in \mathbb{R}^4$$

- Quadrotor dynamic model

$$\dot{p} = Rv$$

$$m\dot{v} = -S(\omega)mv - Te_3 + mgR^T e_3$$

$$\dot{R} = RS(\omega)$$

$$J\dot{\omega} = -S(\omega)J\omega + n$$

- Show that the thrust and rotation matrix can be written as functions of $(y, \dot{y}, \ddot{y}, \dots)$

$$T = f_1(y, \dot{y}, \ddot{y}, \dots) = ? \quad R = f_2(y, \dot{y}, \ddot{y}, \dots) = ?$$

Suggestion: use $m\ddot{p} = -TRe_3 + mge_3$

The quadrotor as a flat system - Exercise

- Circular motion:

$$p(t) = {}^I p_B(t) = r \begin{bmatrix} \cos \psi(t) \\ \sin \psi(t) \\ 0 \end{bmatrix}$$

$$R(t) = {}^I_B R(t) = R_z(\psi(t)), \quad \dot{\psi}(t) = c$$

- Determine the roll and pitch Euler angles

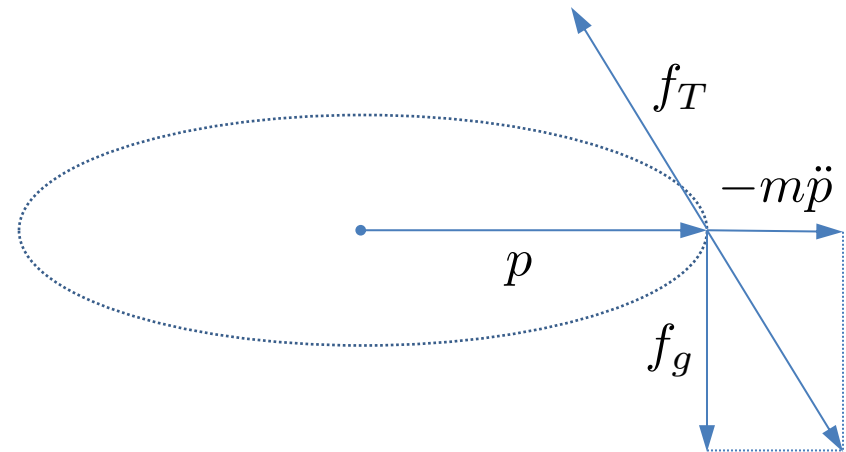
Hint: use

$$m\ddot{p} = f_T + f_g$$

$$f_T = -TRe_3$$

$$f_g = mge_3$$

$$R_y(\theta)R_x(\phi)e_3 = \begin{bmatrix} \cos \phi \sin \theta \\ -\sin \phi \\ \cos \phi \cos \theta \end{bmatrix}$$



Quadrotor Dynamic Modeling - Part II

Design Considerations

Design Considerations

The SWAP (Size, Weight, and Power) tradeoff

- Thrust/Weight Ratio
- Power consumption and Flight time
- Size vs Agility

Design considerations: Thrust/Weight

- Maximum Thrust

- Limited by maximum motor torque $Q_{\max}$

$$T = c_T \omega^2$$

$$|Q| = c_Q \omega^2$$

$$T_{\max} = \frac{c_T}{c_Q} Q_{\max}$$

- Thrust/Weight Ratio

$$T_{\text{hover}} = \frac{1}{4} mg$$

$$\frac{T_{\max}}{T_{\text{hover}}} = \frac{4T_{\max}}{mg} = 4 \frac{c_T}{c_Q} \frac{Q_{\max}}{mg}$$

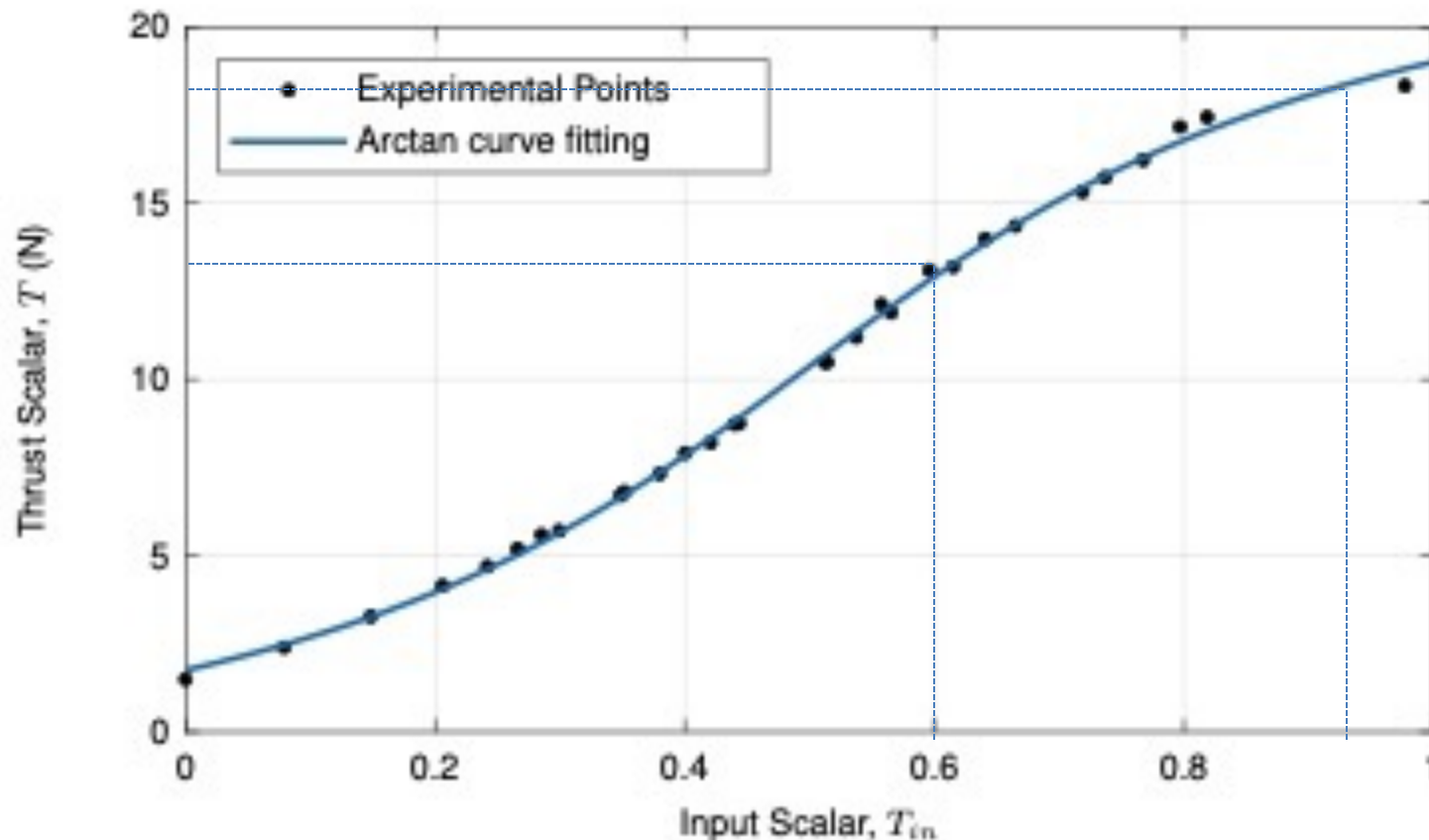
- Typically not greater than 2
- Lower values -> slower response, less control authority

Experimental Thrust Curve Fitting

- Intel Aero drone

- Maximum Thrust to Weight ratio

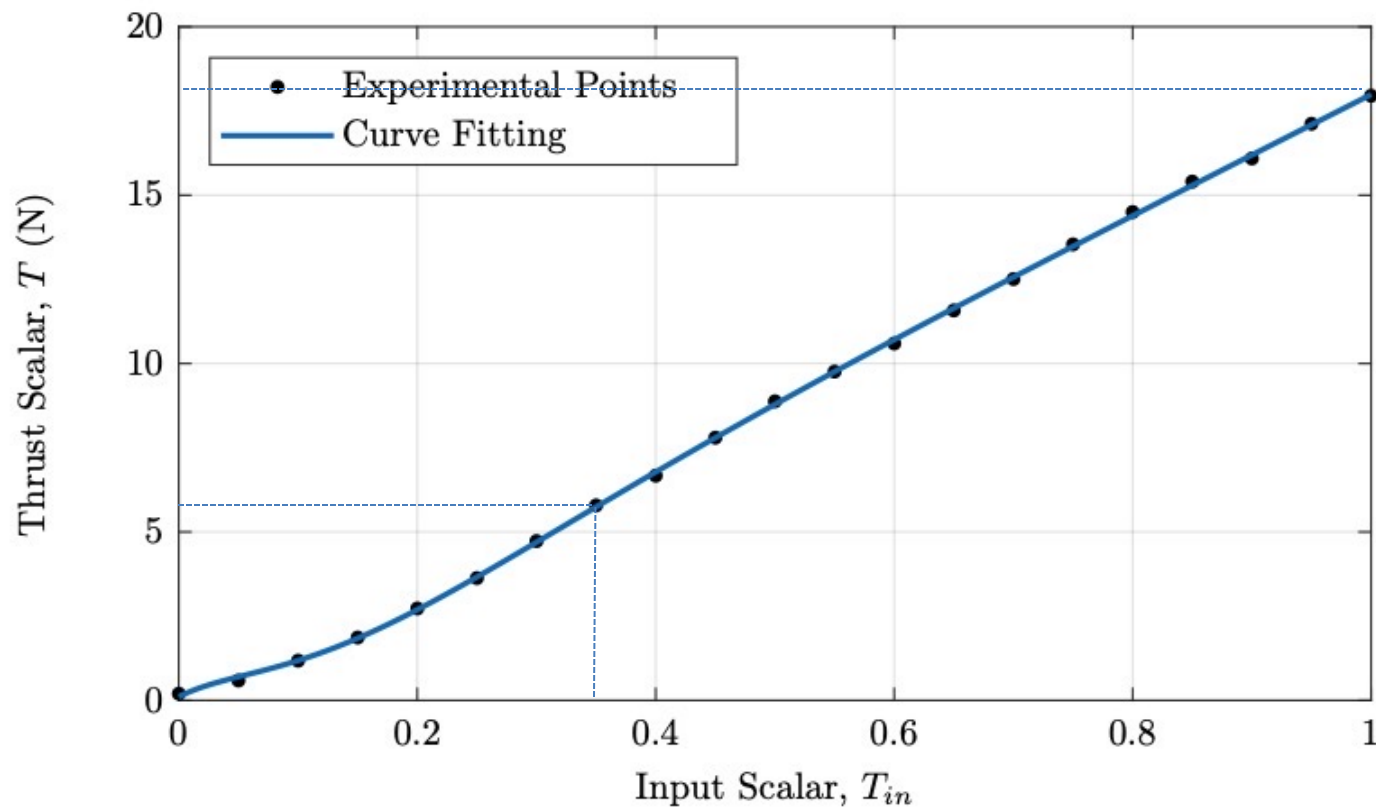
$$\frac{T_{\max}}{T_{\text{hover}}} \approx \frac{18}{mg} = 1.36$$



Experimental Thrust Curve Fitting

- Snapdragon

– Maximum Thrust to Weight ratio $\frac{T_{\max}}{T_{\text{hover}}} \approx \frac{18}{5.6} = 3.2$



Design considerations: flight time

- Mechanical power (for each rotor)

$$T = c_T \omega^2 \quad \omega = \sqrt{\frac{T}{c_T}} \quad p_{motor} = \omega Q = c_Q \omega^3 = c_Q \left(\frac{1}{c_T} T \right)^{3/2}$$

$$|Q| = c_Q \omega^2$$

Mechanical Power proportional to $T^{3/2}$

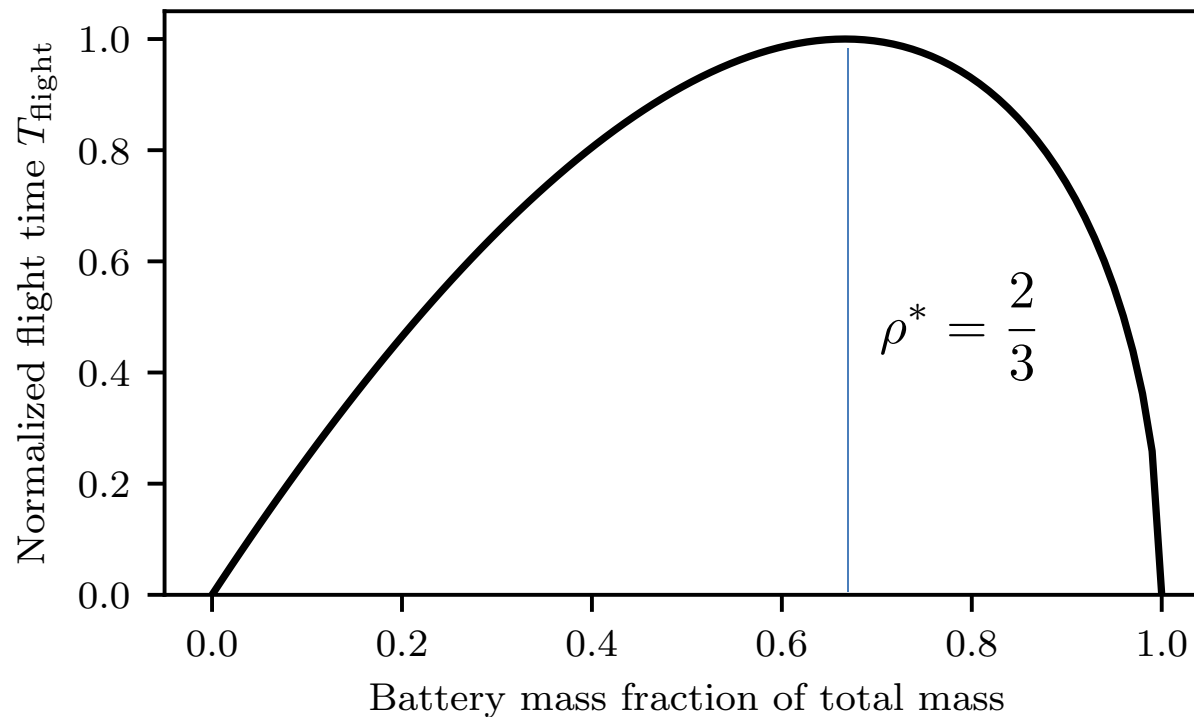
- Maximum flight time

$$t_{flight} = \frac{E_{batt}}{4p_{motor}/\eta + p_{payload}} \quad \eta \approx 70\%$$

- Neglecting payload consumption and assuming hovering condition

$$t_{flight} = \frac{E_{batt}}{p_{hover}} = \frac{E_{batt}}{\frac{c_Q}{2} \left(\frac{1}{c_T} mg \right)^{3/2}} \quad t_{flight} \propto \frac{m_{batt}}{m^{3/2}}$$

Power consumption and battery selection



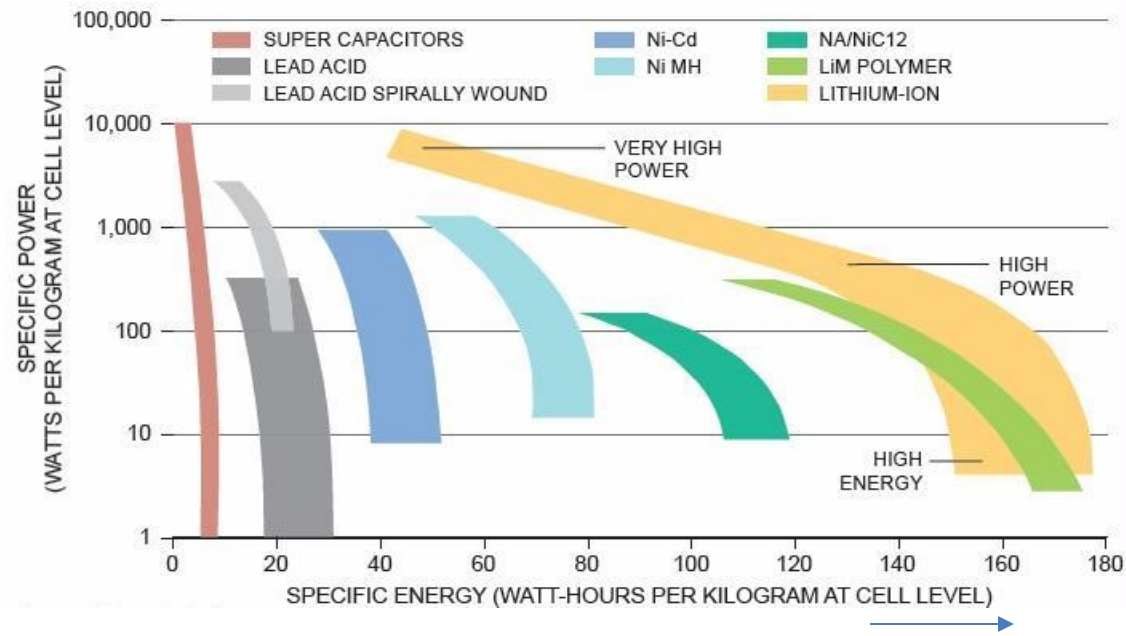
$$t_{flight} \propto \frac{m_{batt}}{m^{3/2}} \quad \rho = \frac{m_{batt}}{m} \Rightarrow t_{flight} \propto \frac{\rho \sqrt{1-\rho}}{m_0}, \quad m_0 = m - m_{batt}$$

Source: <https://spectrum.ieee.org/automaton/robotics/drones/swappable-flying-batteries-keep-drones-aloft-almost-forever>

Design considerations: flight time

- Battery specs
 - Specific power vs specific energy

Increases
Thrust/Weight
ratio



Increases
Flight time



Design considerations: agility

- Size

- Characteristic length d

- Inertia $m \propto d^3, J \propto d^5$

- Rotor speed $r \propto d$

- Mach scaling $\longrightarrow \omega_i \propto \frac{1}{\sqrt{d}}$

Mach Nr: airspeed vs speed of sound

- Froude scaling $\longrightarrow \omega_i \propto \frac{1}{d}$

Foude Nr: airspeed vs gravity

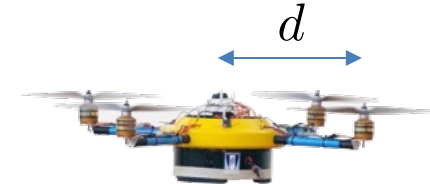
- Angular acceleration

$$\dot{\omega} \propto \frac{Td}{J} \propto \frac{(\omega_i^2 d^4)d}{d^5} = \omega_i^2$$

$\searrow$
 $\swarrow$

$\dot{\omega} \propto \frac{1}{d^2}$
 $\dot{\omega} \propto \frac{1}{d}$

- Conclusion: reducing size increases agility



Source: R. Mahony, V. Kumar and P. Corke, "Multirotor Aerial Vehicles: Modeling, Estimation, and Control of Quadrotor," in IEEE Robotics & Automation Magazine, vol. 19, no. 3, pp. 20-32, Sept. 2012.